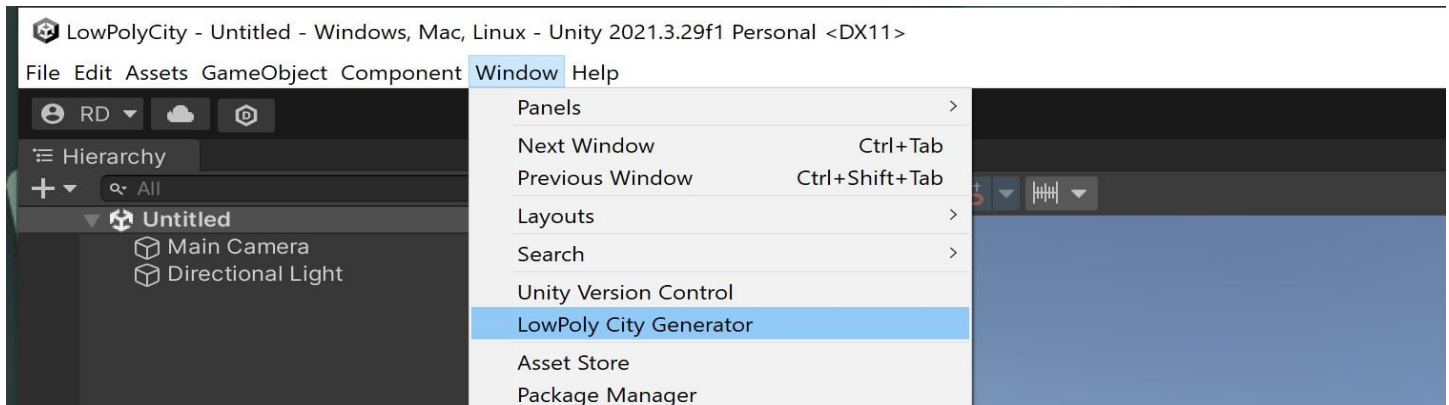


How to use LowPoly City Generator:

1. Import Package
2. Select 'Window | LowPoly City Generator' from the menu

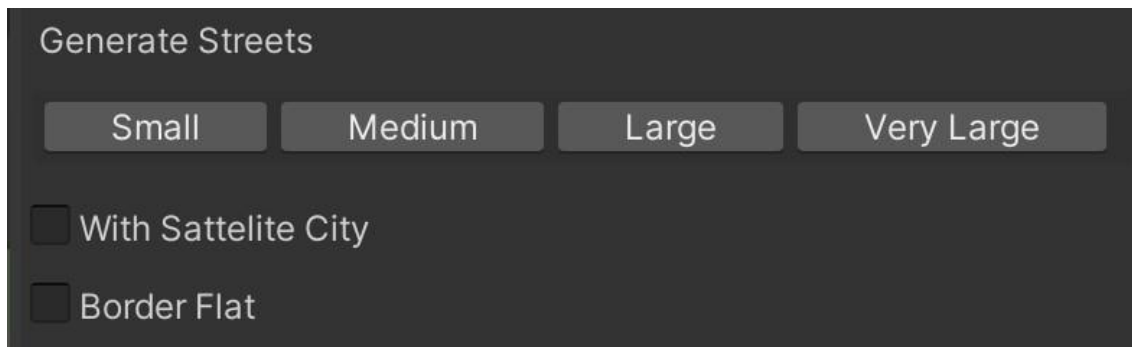


The LowPoly City Generator screen will appear:

A screenshot of the 'LowPoly City Generator' window. It has a title bar 'LP_FCityGenerator'. The main title is 'LowPoly City Generator'. Under 'Generate Streets', there are buttons for 'Small', 'Medium', 'Large', and 'Very Large'. There are checkboxes for 'With Sattelite City' and 'Border Flat'. Under 'Buildings', there are buttons for 'Generate Buildings' and 'Clear Buildings', a checked checkbox for 'With Downtown Area?', and a 'Downtown Size' slider set to 100. Under 'Traffic System', there are buttons for 'Add Traffic System' and 'Remove Traffic System', and checked checkboxes for 'Right Hand' and 'Heavy Traffic'.	Generate Streets <i>Small City - Medium City</i> <i>Large City - Very Large City</i> With Sattelite City Activate if you want to create two cities connected by a Highway Border Flat City with Flat Surroundings, facilitating the insertion of land around the city Buildings <i>Generate Buldings - Clear Buildings</i> With DownTown Area Activate to create a metropolitan city DownTown Size Set the size of the center area Traffic System <i>Add Traffic System</i> <i>Remove Traffic System</i> <i>Right Hand or Left Hand</i> Heavy Traffic
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Opções da tela “LowPoly City Generator”

a) Generate Streets



- Click "**Small City**" to create a small city.
- Click "**Medium City**" to create a medium-sized city.
- Click "**Large City**" to create a large city.
- Click "**Very Large City**" to create a very large city.

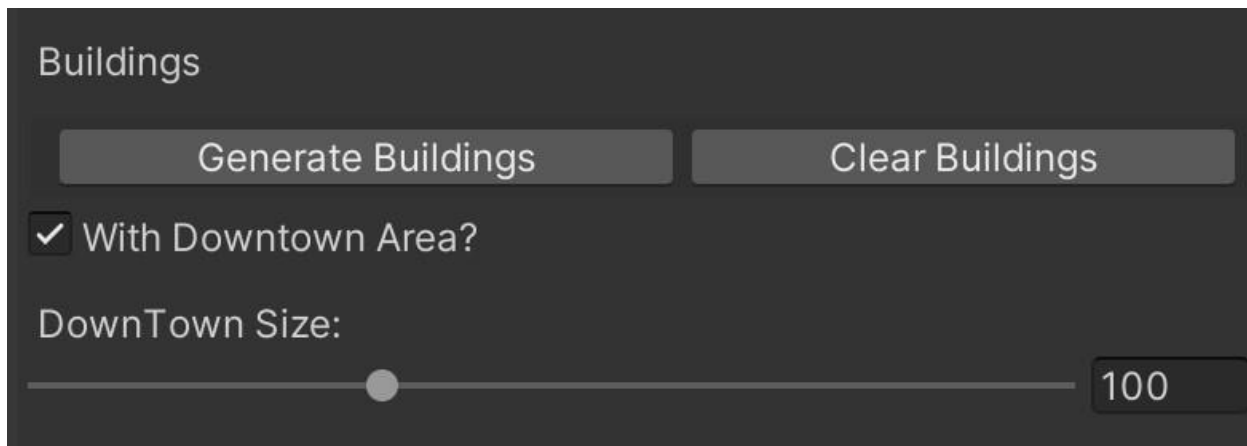
Each time you click one of these buttons, a new city will be generated randomly using street blocks. At this stage, no buildings or other elements (such as parks) are included—only streets.

Every time you generate a city, a new random configuration is created.

Before generating the city, you can adjust the following options:

- **"With Satellite City" Option**
Activate this option to create two cities connected by a highway.
The main city will have the size you selected, and a smaller, secondary city will be generated, connected by a road.
- **"Border Flat" Option**
When enabled, the city will have flat surroundings, making it easier to add terrain around the city.

b) Buildings

A dark-themed control panel for buildings. At the top, the word "Buildings" is displayed. Below it are two buttons: "Generate Buildings" and "Clear Buildings". Under these buttons is a checked checkbox labeled "With Downtown Area?". Below the checkbox is a slider control labeled "DownTown Size:". The slider has a circular handle and a numerical value of "100" displayed at its right end.

This section includes the following buttons:

1. **"Generate Buildings"**

Click to populate the generated city with buildings.

Clicking again will regenerate the buildings randomly, ensuring a unique configuration each time.

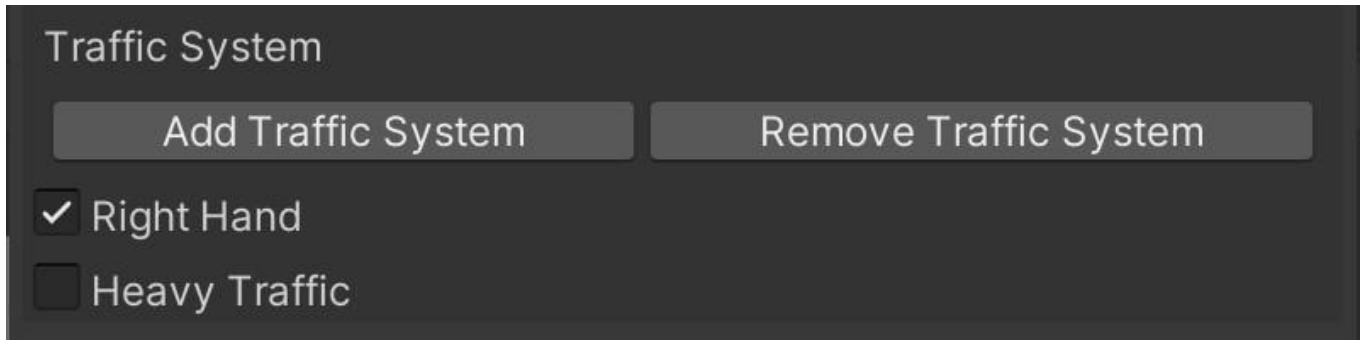
2. **"Clear Buildings"**

Click to remove all buildings, leaving only the streets.

Before clicking **"Generate Buildings"**, you can enable the **"With Downtown Area"** option to create a metropolitan city with a central area containing taller buildings and skyscrapers.

You can also adjust the **downtown size**, ranging from **50 to 200**.

c) Traffic System



1. Click **"Add Traffic System"** to spawn vehicles in the city.

Alternatively, you can manually place the prefab

"Assets/LowPoly City Generator/Traffic System/Traffic System.prefab" into the scene.

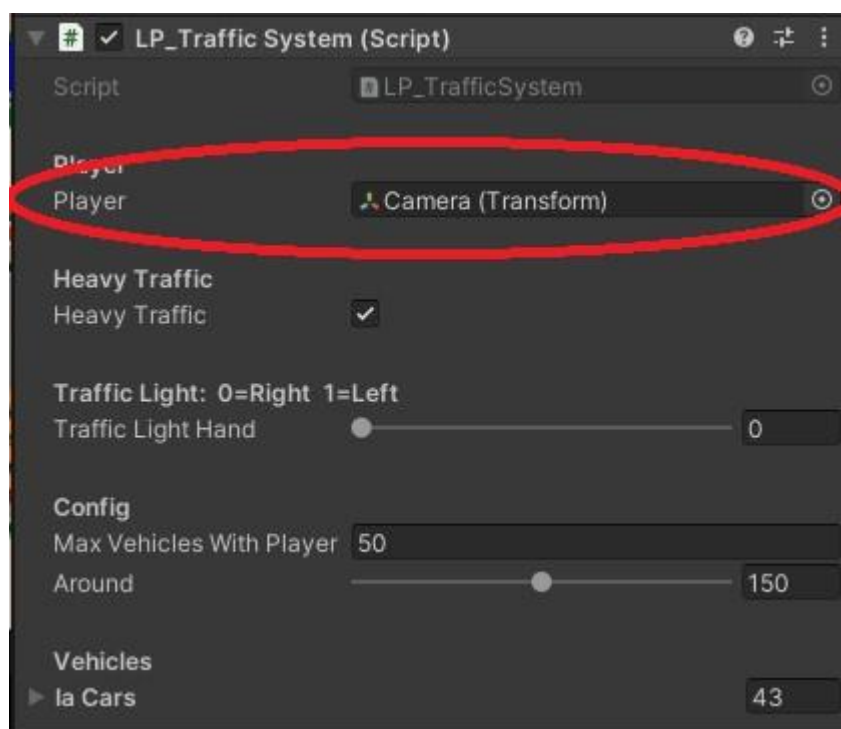
Once added, vehicles will start spawning automatically.

If you prefer vehicles to spawn only during runtime (on Awake), delete the **"CarContainer"** folder from the **Hierarchy**.

2. Player Assignment

- a. Select the **Traffic System** object in the **Hierarchy**.
- b. In the **Inspector**, set the **"Player"** field to the player's camera.

Defining the player is essential for performance optimization.



3. Removing the Traffic System

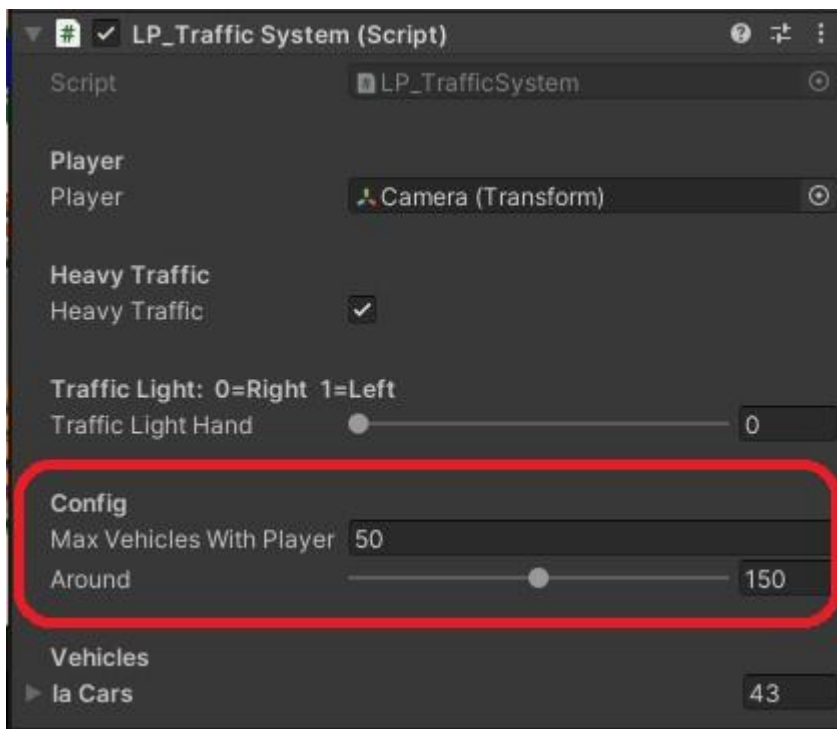
- a. Click **"Remove Traffic System"** or manually delete the **Traffic System** from the **Hierarchy**.

4. Additional Traffic Options

- a. **"Right Hand" Option**: Enables right-hand traffic (disable for left-hand traffic).
- b. **"Heavy Traffic" Option**: Increases the number of vehicles in the scene.

5. Other Traffic Settings

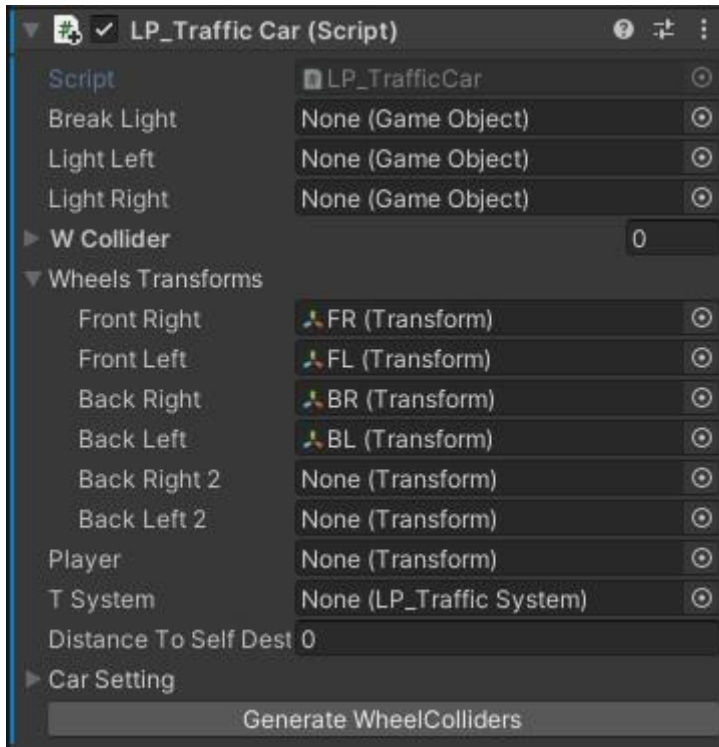
Select the **Traffic System** object in the **Hierarchy**. In the **Inspector**, set:



- a. **"Max Vehicles with Player"**:
Defines the maximum number of vehicles in the scene.
- b. **"Around"**:
Sets the maximum distance at which vehicles will be present relative to the player.

Add Vehicles

You can easily add vehicles to the built-in **Traffic System**.



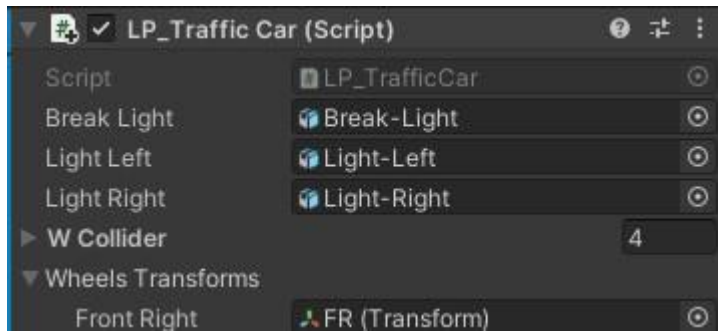
Vehicle requirements:

- The model must have separate wheels.

Steps to add a vehicle:

1. Drag the vehicle model into the scene.
2. In the **Hierarchy**, select the model and in the **Inspector**, click "**Add Component**", then add "**LP_TrafficCar**".
3. Assign the wheel objects to their respective fields:
 - a. **Front Left** → Assign the front left wheel.
 - b. **Front Right** → Assign the front right wheel.
 - c. **Back Left** → Assign the rear left wheel.
 - d. **Back Right** → Assign the rear right wheel.
4. Click "**Generate WheelColliders**" to automatically create the **WheelColliders** and **Rigidbody**.
5. **Ensure the vehicle has a collider!**
 - a. Add a **Mesh Collider** or **Box Collider** to the car's body.

Adding Brake Lights and Turn Signals:



- **Brake Light** → This object will activate when braking.
- **Left Turn Signal** → This object will flash when turning left.
- **Right Turn Signal** → This object will flash when turning right.

Each of these can be a **Light component** (orange for turn signals) or a **mesh with emissive material**.



Video tutorial available at:

<https://youtu.be/kNI5r20oQAQ>

Using Third-Party Traffic Systems

If you prefer to use a different traffic system, simply avoid adding the built-in Traffic System prefab. You can then implement your preferred system.

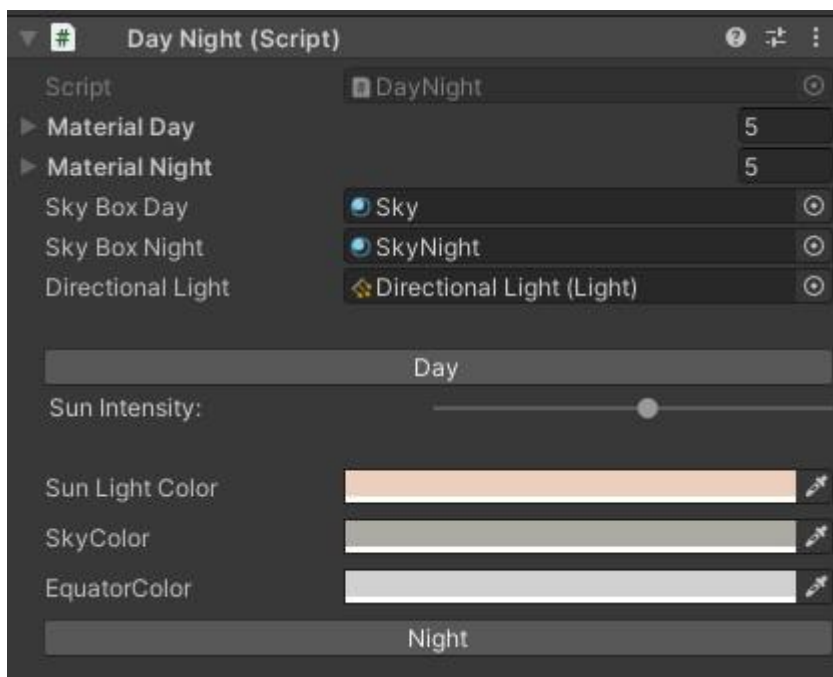
If you're not using the included traffic system, it is recommended to delete all "**TS**" folders inside "**LP-City**" in the **Hierarchy**. These folders contain the waypoints used by the builtin system.

Day / Night Scenes

Adding Night/Day Scene option

- 1) Create a city using the LPCG
- 2) Add the prefab "LowPoly City Generator/DayNight/DayNight" to the scene
- 3) Select the DayNight in the Hierarchy, and in the Inspector (in the Directional Light field) set the Directional light of the scene.
- 4) Click 'Day' button for daytime scene or 'Night' button for night scene.

You will find settings to control the appearance of both day and night.



Switch Day/Night at runtime

The code to switch day and night is this:

```
DayNight dayNight = FindObjectOfType<DayNight>();  
if (dayNight)  
{  
    dayNight.isNight = !dayNight.isNight; // true or false  
    dayNight.ChangeMaterial();  
}
```

Example code and scene:

In *Assets/LowPoly City Generator/DayNight*, you will find a script named **"ShiftAtRuntime.cs"**.

- 1) Create a city using the LPCG
- 2) Add the prefab "LowPoly City Generator/DayNight/DayNight" to the scene
- 3) Select the DayNight in the Hierarchy, and in the Inspector (in the Directional Light field) set the Directional light of the scene.
- 4) Click 'Day' button for daytime scene or 'Night' button for night scene.
- 5) Attach the script **"ShiftAtRuntime.cs"** to an object in the scene.

By doing this, you will be able to switch between day and night during runtime by pressing the **"N"** key.

Check the **"ShiftAtRuntime.cs"** script to learn more.

You can also open the example scene "LowPoly City Generator/Scenes/DayNight Runtime Demo".

Video: Switch day/night in one click:

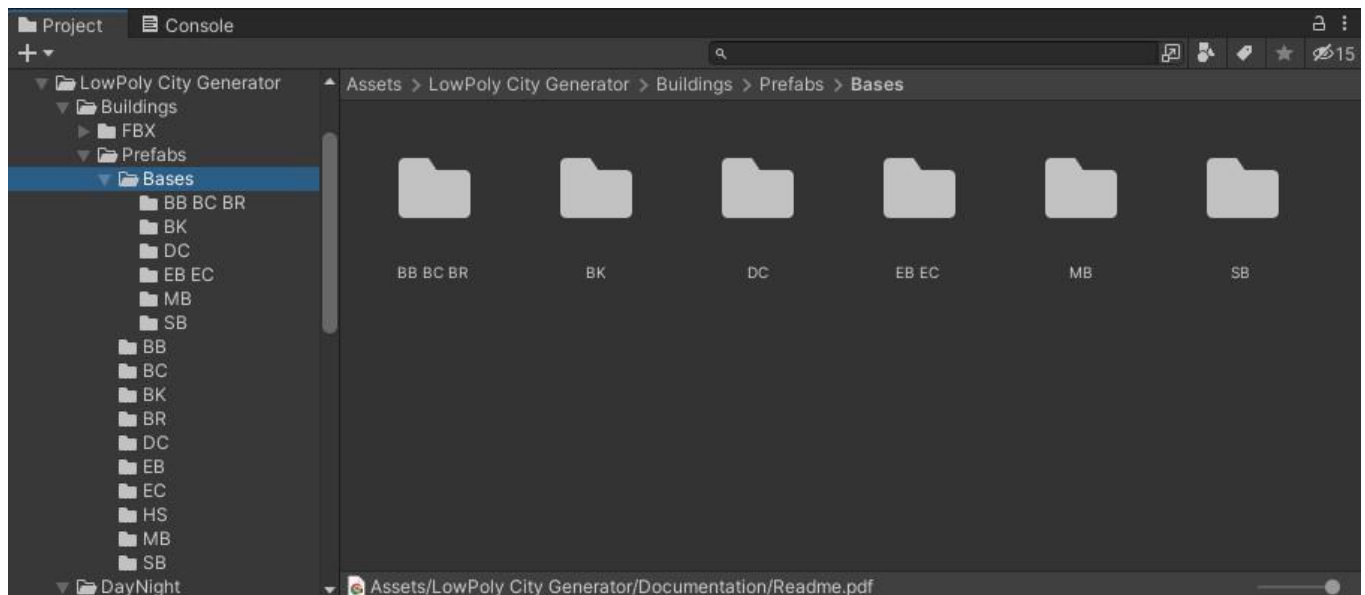
<https://youtu.be/7NtpdIZ66Ug>

Add third-party Buildings

You can add your own or third-party buildings to the city.

The **LowPoly City Generator** categorizes buildings by type. Below is a list of the available building types:

BC	Downtown street building (not on a corner)
EB	Corner buildings in suburban areas
BB	Buildings in suburban areas (not on a corner)
BR	Residential buildings in suburban areas (not on a corner)
HS	Houses in suburban areas (not on a corner)
BK	Buildings that occupy an entire block
SB	Large buildings that occupy bigger blocks
DC	Corner buildings that occupy one side of the street
MB	Buildings that occupy both sides of the street



Steps to add a third-party building:

1. Choose the correct building base

- a. The ground where buildings are placed is called a **base**.
- b. In "**Assets/LowPoly City Generator/Buildings/Prefabs/Bases**", you'll find bases categorized by building type.
- c. Select the appropriate base and place it in the scene.

2. Place the building

- a. Drag your building model into the scene.
- b. In the **Hierarchy**, place the building inside the corresponding base object.
- c. Rename the base object to match the building name.

3. Save the building as a prefab

- a. In "**Assets/LowPoly City Generator/Buildings/Prefabs/**", you'll find folders labeled **BB**, **BC**, **EB**, etc.
- b. These folders store prefabs for different building types.
- c. Save your building prefab in the appropriate folder.
- d. Once saved, **LowPoly City Generator** will automatically use the newly added building in future city generations.

A video tutorial for this process is available at:

<https://youtu.be/SAAz4kqh9jc>

Adding Third Party Buildings to the DayNight System

If your third-party buildings have day and night versions of their materials, you can integrate them into the **Day/Night System** so they switch correctly.

Steps to set up day/night materials:

1. Select **DayNight** in the **Hierarchy**.
2. In the **Inspector**, locate:
 - a. "**MaterialDay**": Add the daytime version of the materials.
 - b. "**MaterialNight**": Add the nighttime version of the materials.

Important:

Only materials that have separate day and night versions need to be included. Materials that remain unchanged do not need to be added.

The **index order** in **MaterialDay** must match the **index order** in **MaterialNight**.

Example:

- When switching to night mode, **MaterialDay[1]** will be replaced by **MaterialNight[1]**.

Adding Third-Party Props

Cities are generated using street and border prefabs located in:

📁 "Assets/LowPoly City Generator/Roads/Prefabs/Blocks"

📁 "Assets/LowPoly City Generator/Roads/Prefabs/Border"

You can edit these prefabs to add or replace elements.

For example, replace trees, add street signs, lampposts, cones, or any other objects.

⚠️ **Be careful not to place props where buildings or roads should be, as this may obstruct vehicle movement.**

Support:

support@masterpixel3d.com